

- 2 (a) An Olympic diver stands on a platform above a pool of water, as shown in Fig. 2.1.

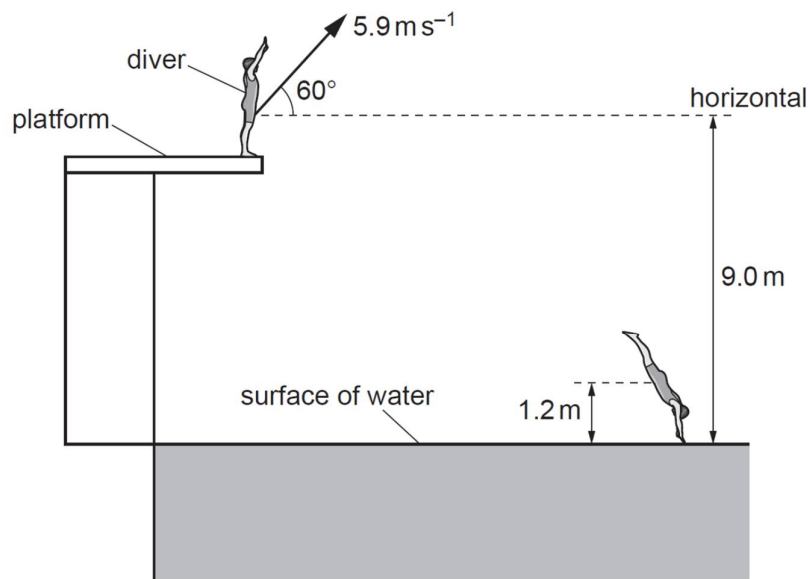


Fig. 2.1 (not to scale)

When the diver is on the platform his centre of gravity is a vertical height of 9.0 m above the surface of the water. The diver jumps from the platform with a velocity of 5.9 m s^{-1} at an angle of 60° to the horizontal.

Air resistance is negligible.

When the diver hits the surface of the water, his centre of gravity is a vertical height of 1.2 m above the surface of the water.

- (i) Calculate the time it takes him to hit the surface of the water.

time = s [2]

- (ii) Calculate the angle from the water surface when he hits the surface of the water.

angle = ° [3]

- (b) The diver in (a) enters the water.

- (i) Explain the why the resultant force acting on the diver in decreases in the water as he moves downwards.

.....

 [2]

- (ii) The diver has a volume of $7.5 \times 10^{-2} \text{ m}^3$. The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Show that the upthrust acting on the diver when he is entirely underwater is 740 N.

[1]

- (iii) At a particular instant when the diver is entirely underwater his horizontal velocity is zero. The viscous drag force acting on him at this instant is 950 N vertically upwards. The diver has a mass of 78 kg.

Determine the magnitude and direction of the acceleration of the diver.

acceleration = m s^{-2}

direction

[3]

[Total: 11]